Mini Project

Implementing a Mobile Communication System

Objective

This project is designed to help students understand object-oriented programming principles by modeling a simple mobile communication system. Students will implement a set of classes representing the essential components of a phone communication network. This system will simulate the requirements for a phone to initiate and receive calls within a network area covered by a set of antennas.

Project Description

In this project, students will create a system that includes classes for **Phone**, **Antenna**, **SimCard**, and **Network**. Each class will have specific attributes and methods, modeling the conditions required for a phone to communicate within a defined network.

Each student is expected to analyze the system requirements and determine the appropriate attributes and methods for each class. The project aims to encourage critical thinking in designing class structures, relationships, and functional methods.

System Requirements

1. For a Phone to make a call, it must:
    - Be charged (have sufficient battery level).
    - Have an active SIM card.
    - Have sufficient credit on its SIM card.
    - Be within the network's coverage range (or within at least in the range of one antenna).
    - The phone connects to the nearest antenna based on its location.
    - Ensure that the antenna it connects to has **available capacity** (the number of active communications does not exceed its limit).
    - The call is immediately disconnected when the phone moves out of the network's coverage range (e.g., if the person is walking while on the call).
2. For a **Phone** to **receive a call**, it must:
    - Be **charged**.
    - Be **within the range** of an antenna.

- The caller's phone and the receiver's phone do not need to be connected to the same antenna.

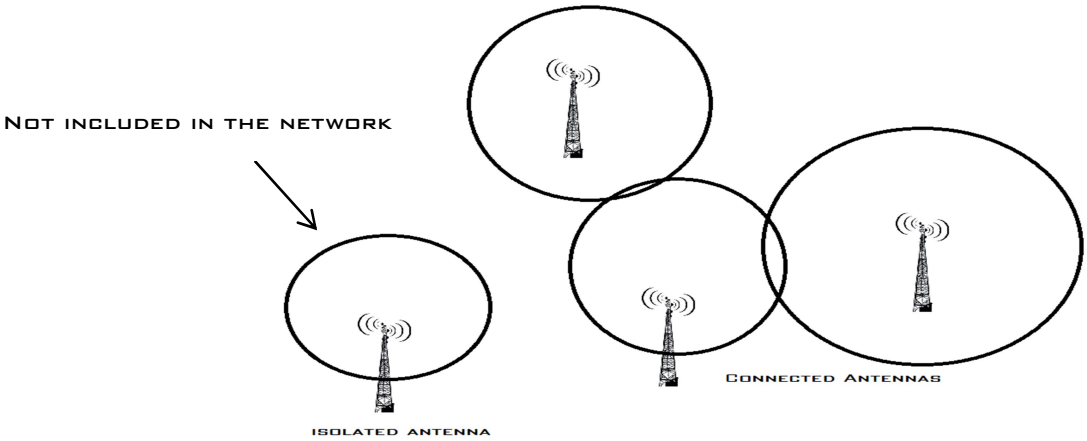
Class Descriptions

1. **Phone Class**:
   - Represents a mobile device that can initiate and receive calls, and change position.
   - Requirements: Students need to define attributes to:
     - Track the **battery level**.
     - Link a **SimCard** to the phone.
     - Represent the **location** of the phone.
   - Methods: Implement methods to:
     - Check if the phone can make or receive calls based on the system requirements.
     - Once a phone is connected to a call, it will reject all incoming calls for the duration of the current call.
     - The call method should handle two scenarios: a stationary call (no position change) and a moving call (phone's position change during the call).
2. **SimCard Class**:
   - Represents the SIM card mounted in a phone.
   - Requirements: Define attributes to:
     - Give a fixed phone number for the Sim Card.
     - Once activated, a Sim Card can communicate.
     - Track the **credit balance** of the SIM card.
   - Methods: Create methods to:
     - Deduct credit when making a call.
     - Check if there is sufficient credit available for a call.
3. **Antenna Class**:
   - Represents a cellular antenna that provides network coverage within a defined area.
   - Requirements: Define attributes to:
     - Represent the **location** of the antenna.
     - Define a **coverage radius** specifying the area where the antenna provides network access.
     - Set a **communication capacity limit** specifying the maximum number of active calls allowed simultaneously.

- Track the **current number of active communications** on the antenna.

- Methods: Implement methods to:
  - Check if a given phone location is within the coverage area.
  - Track active communications, including:
    - **incrementActiveCalls()**: Increases the number of active calls if it does not exceed the capacity limit.
    - **decrementActiveCalls()**: Decreases the count when a call ends.
    - **canAcceptNewCall()**: Returns `true` if the number of active communications is below the capacity limit, allowing an additional call to be established.

4. **Network Class**:
   - Represents a mobile network composed of multiple antennas that cover a larger area.
   - Requirements: Define attributes to:
     - Store a list of **Antennas** in the network.
     - Implementation of the **Network** class should consider the constraint that only antennas connected to at least one other antenna's coverage area can be added. Isolated antennas, which do not overlap with the coverage area of any existing antenna in the network, are not allowed.



   - Methods: Include methods to:
     - Add antennas to the network.
     - Find the nearest antenna to a given phone location, check if it is within range, and ensure that the antenna has capacity for an additional communication.

Instructions

- **Step 1**: Analyze the requirements and decide on appropriate attributes for each class.
- **Step 2**: Implement each class according to the descriptions, including necessary methods to enforce communication rules.
- **Step 3**: Create a main program or test cases to simulate:
  - Various scenarios of making and receiving calls based on battery level, credit balance, network coverage, mobility…
  - Edge cases, such as an antenna reaching its maximum communication capacity or a phone being out of range.

Submission Details

- **Submission Deadline**: The project must be submitted by the end of the **The end of semeter**
- **Group Size**: Each group can consist of up to a maximum of **4 students**.
- **Code Quality Requirements**:
  - The source code must be **well-organized** and follow clean coding practices.
  - All classes should include **constructors**, as well as **getters and setters** for the attributes.
  - The code must be thoroughly **commented**, explaining the purpose of methods, attributes, and key logic implementations.
- To send by email in one compressed file to:

  Group 1: **nasri.khaled@univ-setif.dz**